

[Iron: Fact Sheet for Consumers \(original version\)](#)

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What is iron and what does it do?

Iron is a mineral that the body needs for growth and development. Your body uses iron to make hemoglobin, a protein in red blood cells that carries oxygen from the lungs to all parts of the body, and myoglobin, a protein that provides oxygen to muscles. Your body also needs iron to make some hormones.

How much iron do I need?

The amount of iron you need each day depends on your age, your sex, and whether you consume a mostly plant-based diet. Average daily recommended amounts are listed below in milligrams (mg). Vegetarians who do not eat meat, poultry, or seafood need almost twice as much iron as listed in the table because the body doesn't absorb nonheme iron in plant foods as well as heme iron in animal foods.

Life Stage	Recommended Amount
Birth to 6 months	0.27 mg
Infants 7–12 months	11 mg
Children 1–3 years	7 mg
Children 4–8 years	10 mg
Children 9–13 years	8 mg
Teen boys 14–18 years	11 mg
Teen girls 14–18 years	15 mg
Adult men 19–50 years	8 mg
Adult women 19–50 years	18 mg
Adults 51 years and older	8 mg
Pregnant teens	27 mg
Pregnant women	27 mg
Breastfeeding teens	10 mg
Breastfeeding women	9 mg

What foods provide iron?

Iron is found naturally in many foods and is added to some fortified food products. You can get recommended amounts of iron by eating a variety of foods, including the following:

- Lean meat, seafood, and poultry
- Iron-fortified breakfast cereals and breads
- White beans, lentils, spinach, kidney beans, and peas
- Nuts and some dried fruits, such as raisins

Iron in food comes in two forms: heme iron and nonheme iron. Nonheme iron is found in plant foods and iron-fortified food products. Meat, seafood, and poultry have both heme and nonheme iron.

Your body absorbs iron from plant sources better when you eat it with meat, poultry, seafood, and foods that contain vitamin C, such as citrus fruits, strawberries, sweet peppers, tomatoes, and broccoli.

What kinds of iron dietary supplements are available?

Iron is available in many multivitamin/mineral supplements and in supplements that contain only iron. Iron in supplements is often in the form of ferrous sulfate, ferrous gluconate, ferric citrate, or ferric sulfate. Dietary supplements that contain iron have a statement on the label warning that they should be kept out of the reach of children. Accidental overdose of iron-containing products is a leading cause of fatal poisoning in children under 6.

Am I getting enough iron?

Most people in the United States get enough iron. However, certain groups of people are more likely than others to have trouble getting enough iron:

- Teen girls and women with heavy periods
- Pregnant women and teens
- Infants (especially if they are premature or low-birth weight)
- Frequent blood donors
- People with cancer, gastrointestinal (GI) disorders, or heart failure

What happens if I don't get enough iron?

In the short term, getting too little iron does not cause obvious symptoms. The body uses its stored iron in the muscles, liver, spleen, and bone marrow. However, when levels of iron stored in the body become low, iron deficiency anemia sets in. Red blood cells become smaller and contain less hemoglobin. As a result, blood carries less oxygen from the lungs throughout the body.

Symptoms of iron deficiency anemia include GI upset, weakness, tiredness, lack of energy, and problems with concentration and memory. In addition, people with iron deficiency anemia are less able to fight off germs and infections, to work and exercise, and to control their body temperature. Infants and children with iron deficiency anemia might develop learning difficulties.

Iron deficiency is not uncommon in the United States, especially among young children, women under 50, and pregnant women. It can also occur in people who do not eat meat, poultry, or seafood; lose blood; have GI diseases that interfere with nutrient absorption; or eat poor diets.

What are some effects of iron on health?

Scientists are studying iron to understand how it affects health. Iron's most important contribution to health is preventing iron deficiency anemia and resulting problems.

Pregnant women

During pregnancy, the amount of blood in a woman's body increases, so she needs more iron for herself and her growing baby. Getting too little iron during pregnancy increases a woman's risk of iron deficiency anemia and her infant's risk of low birth weight, premature birth, and low levels of iron. Getting too little iron might also harm her infant’s brain development.

Women who are pregnant should talk with their health care provider and take an iron supplement if recommended.

Infants and toddlers

Iron deficiency anemia in infancy can lead to delayed psychological development, social withdrawal, and less ability to pay attention. By age 6 to 9 months, full-term infants could become iron deficient unless they eat iron-enriched solid foods or drink iron-fortified formula.

Anemia of chronic disease

Some chronic diseases—such as rheumatoid arthritis, inflammatory bowel disease, and some types of cancer—can interfere with the body's ability to use its stored iron. Taking more iron from foods or supplements usually does not reduce the resulting anemia of chronic disease because iron is diverted from the blood circulation to storage sites. The main therapy for anemia of chronic disease is treatment of the underlying disease.

Can iron be harmful?

Yes, iron can be harmful if you get too much. In healthy people, taking high doses of iron supplements (especially on an empty stomach) can cause an upset stomach, constipation, nausea, abdominal pain, vomiting, and diarrhea. Large amounts of iron might also cause more serious effects, including inflammation of the stomach lining and ulcers. High doses of iron can also decrease zinc absorption. Extremely high doses of iron (in the hundreds or thousands of mg) can cause organ failure, coma, convulsions, and death. Child-proof packaging and warning labels on iron supplements have greatly reduced the number of accidental iron poisonings in children.

Some people have an inherited condition called hemochromatosis that causes toxic levels of iron to build up in their bodies. Without medical treatment, people with hereditary hemochromatosis can develop serious problems such as liver cirrhosis, liver cancer, and heart disease. People with this disorder should avoid using iron supplements and vitamin C supplements.

The daily upper limits for iron include intakes from all sources—food, beverages, and supplements—and are listed below. A doctor might prescribe more than the upper limit of iron to people who need higher doses for a while to treat iron deficiency.

Ages	Upper Limit
Birth to 12 months	40 mg
Children 1–13 years	40 mg
Teens 14–18 years	45 mg
Adults 19+ years	45 mg

Does iron interact with medications or other dietary supplements?

Yes, iron supplements can interact or interfere with medicines and other supplements you take. Here are several examples:

- Iron supplements can reduce the amount of levodopa that the body absorbs, making it less effective. Levodopa, found in Sinemet and Stalevo, is used to treat Parkinson's disease and restless leg syndrome.
- Taking iron with levothyroxine can reduce this medication's effectiveness. Levothyroxine (Levothroid, Levoxyl, Synthroid, Tirosint, and Unithroid) is used to treat hypothyroidism, goiter, and thyroid cancer.
- The proton pump inhibitors lansoprazole (Prevacid) and omeprazole (Prilosec) decrease stomach acid, so they might reduce the amount of nonheme iron that the body absorbs from food.
- Calcium might interfere with iron absorption. Taking calcium and iron supplements at different times of the day might prevent this problem.

Tell your doctor, pharmacist, and other health care providers about any dietary supplements and prescription or over-the-counter medicines you take. They can tell you if the dietary supplements might interact with your medicines or if the medicines might interfere with how your body absorbs, uses, or breaks down nutrients.

Iron and healthful eating

People should get most of their nutrients from food and beverages, according to the federal government's *Dietary Guidelines for Americans*. Foods contain vitamins, minerals, dietary fiber, and other components that benefit health. In some cases, fortified foods and dietary supplements are useful when it is not possible to meet needs for one or more nutrients (for example, during specific life stages such as pregnancy). For more information about building a healthy dietary pattern, see the [Dietary Guidelines for Americans](#).

Where can I find out more about iron?

- For general information on iron
 - Office of Dietary Supplements (ODS) Health Professional Fact Sheet on [Iron](#)
 - MedlinePlus, [Iron](#) and [Iron in diet](#)
- For more information on food sources of iron
 - USDA's [FoodData Central](#)
 - USDA, Nutrient List for Iron (listed by [food](#) or by [iron content](#))
- For more advice on choosing dietary supplements
 - ODS [Frequently Asked Questions: Which brand\(s\) of dietary supplements should I purchase?](#)
- For information about building a healthy dietary pattern
 - [Dietary Guidelines for Americans](#)

Glossary

absorption

In nutrition, the process of moving protein, carbohydrates, fats, and other nutrients from the digestive system into the bloodstream. Most absorption occurs in the small intestine.

anemia

A condition in which the number of red blood cells in the blood, or the amount of hemoglobin in them, is lower than normal, causing a condition in which red blood cells are not able to supply enough oxygen to all the tissues in the body. Hemoglobin is the substance in red blood cells that carries oxygen to your body's cells.

bone marrow

The soft, sponge-like tissue in the center of most bones. It produces white blood cells, red blood cells, and platelets.

calcium

A mineral found throughout the body. Calcium is needed for healthy bones and teeth, for nerves and enzymes to function properly, and for blood clotting. Calcium is found in some foods, including milk, yogurt, and cheese, and in Chinese cabbage, kale, broccoli and fortified foods, such as many drinks, tofu, and cereals.

cancer

A group of diseases in which cells divide abnormally and without control, and spread to nearby tissues and other parts of the body. Without treatment, cancer can stop organs from working normally, damage

body systems, and cause the patient to die. Cancer may be caused by multiple factors, such as radiation, sunlight, tobacco, certain viruses, and poisonous chemicals; however, the cause of many cancers is unknown.

cell

The individual unit that makes up the tissues of the body. All living things are made up of one or more cells, which are the smallest units of living structure capable of independent existence.

chronic

Happening for a long time, persistently, or repeatedly.

chronic disease

A condition that is continuous or recurrent, is not easily cured, and cannot be passed from person to person. Examples of chronic diseases include heart disease, diabetes, and asthma.

cirrhosis

A condition in which damaged liver cells are replaced with scar tissue, making it progressively more difficult for the liver to function properly.

constipation

A condition in which stool becomes hard, dry, and difficult to pass and bowel movements happen infrequently. Other symptoms may include painful bowel movements and feeling bloated, uncomfortable, and sluggish.

consume

To eat or drink.

control

In a clinical trial, the group of participants that does not receive the new treatment being studied. This group is compared with the group receiving the new treatment, to see whether the new treatment works. In an observational study, the controls are participants who do not have a particular health condition; the control group is compared with the group of participants who do have the condition to see if certain factors (such as diet, activity level, or use of dietary supplements) may be associated with developing or preventing the condition.

deficiency

An amount that is not enough; a shortage.

diarrhea

Loose, watery stools.

dietary fiber

A substance in plants that you cannot digest. It adds bulk to your diet to make you feel full, helps prevent constipation, and may help lower the risk of heart disease and diabetes. Good sources of dietary fiber include whole grains (such as brown rice, oats, quinoa, bulgur, and popcorn), legumes (such as black beans, garbanzo beans, split peas, and lentils), nuts, seeds, fruit, and vegetables.

Dietary Guidelines for Americans

Advice from the federal government to promote health and reduce the chance (risk) of long-lasting (chronic) diseases through nutrition and physical activity. The Guidelines are updated and published every 5 years by the US Department of Health and Human Services and the US Department of Agriculture.

dietary supplement

A product that is intended to supplement the diet. A dietary supplement contains one or more dietary ingredients (including vitamins, minerals, herbs or other botanicals, amino acids, and other substances) or their components; is intended to be taken by mouth as a pill, capsule, tablet, or liquid; and is identified on the front label of the product as being a dietary supplement.

disorder

In medicine, a disturbance of normal functioning of the mind or body. Disorders may be caused by genetic factors, disease, or trauma.

dose

The amount of medicine or other substance taken at one time or over a specific period of time.

effectiveness

In medicine, the ability of an intervention (for example, a drug, surgery, or a dietary supplement) to produce the desired beneficial effect under the usual conditions of care by a health care provider.

enriched

When certain nutrients (such as vitamins and minerals) have been added to a food product to replace nutrients that may be lost during processing or storage. For example, white flour is enriched with

thiamin, riboflavin, niacin, and iron because those nutrients are lost when grain is made into flour. Folic acid is also added to enriched flour.

fatal

Deadly; causing death.

fortified

When nutrients (such as vitamins and minerals) are added to a food product. For example, when calcium is added to orange juice, the orange juice is said to be "fortified with calcium". Similarly, many breakfast cereals are "fortified" with several vitamins and minerals.

gastrointestinal

GI. Having to do with the gastrointestinal tract (the large, muscular tube that extends from the mouth to the anus, where the movement of muscles and release of hormones and enzymes digest food).

goiter

An enlarged thyroid gland. A goiter is caused by too little iodine in the diet or by other conditions, such as a growth on the thyroid or a gland that makes too much or not enough hormones.

heart failure

A condition in which the heart is unable to pump the amount of blood needed by the body. It is caused by high blood pressure, heart attack, and other disorders of the heart or blood vessels. Also called congestive heart failure.

hemochromatosis

A condition in which the body absorbs more iron than it needs and stores it in the liver, heart, and pancreas. Hemochromatosis causes liver disease, heart problems, and organ failure.

hemoglobin

The substance inside red blood cells that carries oxygen from the lungs to the body's tissues.

hormone

A group of chemicals made by glands in the body. Hormones circulate in the bloodstream and control the actions of certain cells or organs. Some hormones can also be manufactured.

hypothyroidism

A disorder in which the thyroid gland makes too little hormone for the body to function well. Thyroid hormones affect chemical reactions in the body, brain development, breathing, heart and nervous system functions, body temperature, muscle strength, skin dryness, menstrual cycles, cholesterol levels, and body weight.

infant

A child younger than 12 months old.

infection

The invasion and spread of germs in the body. The germs may be bacteria, viruses, yeast, or fungi.

inflammation

Redness, swelling, pain, and/or a feeling of heat in an area of the body. It is a protective reaction to injury, disease, or irritation of tissues.

inflammatory bowel disease

IBD. Long-lasting (chronic) problems that cause irritation and ulcers in the digestive tract. The most common disorders are ulcerative colitis and Crohn's disease.

inherit

In genetics, to receive genes that are passed from parents to their children.

interaction

A change in the way a dietary supplement acts in the body when taken with certain other supplements, medicines, or foods, or when taken with certain medical conditions. Interactions may cause the dietary supplement to be more or less effective, or cause effects on the body that are not expected.

iron

In nutrition, a mineral the body needs to make red blood cells, proteins, and enzymes; and for the control of cell growth and cell specialization. Iron is found in some foods, including red meats, fish, poultry, lentils, and beans.

kidney

One of two organs that remove waste from the blood (as urine). The kidneys also make erythropoietin (a substance that stimulates red blood cell production) and help regulate blood pressure. The kidneys are located near the back under the lower ribs.

label

When referring to dietary supplements, information that appears on the product container, including a descriptive name of the product stating that it is a "supplement"; the name and place of business of the

manufacturer, packer, or distributor; a complete list of ingredients; and each dietary ingredient contained in the product. Supplements must also include directions for use, nutrition labeling in the form of a Supplement Facts panel that identifies each dietary ingredient contained in the product and the serving size, amount, and active ingredients.

liver

A large organ located in the right upper abdomen. It stores nutrients that come from food, makes chemicals needed by the body, and breaks down some medicines and harmful substances so they can be removed from the body.

lung

An organ in the chest that supplies oxygen to the body and removes carbon dioxide. There are two lungs in the body.

milligram

mg. A measure of weight. It is a metric unit of mass equal to 0.001 gram (it weighs 28,000 times less than an ounce).

mineral

In nutrition, an inorganic substance found in the earth that is required to maintain health.

nausea

The uneasy feeling of having an urge to throw up (vomit).

nutrient

A chemical compound in food that is used by the body to function and maintain health. Examples of nutrients include proteins, fats, carbohydrates, vitamins, and minerals.

Office of Dietary Supplements

ODS, Office of Disease Prevention, Office of Director, National Institutes of Health, Department of Health and Human Services. ODS strengthens knowledge and understanding of dietary supplements by evaluating scientific information, stimulating and supporting research, disseminating research results, and educating the public to foster an enhanced quality of life and health for the US population.

Parkinson's disease

A disease in older people that causes nervous system symptoms that worsen slowly over time. Symptoms include trembling hands, arms, legs, and face; stiff arms, legs, and torso; slow movements; and impaired balance and coordination. Parkinson's disease can also affect emotions, memory, judgment, and reasoning.

pharmacist

A person licensed to make and dispense (give out) prescription drugs and who has been taught how they work, how to use them, and their side effects.

poultry

Birds that are raised for eggs or meat, including chickens, turkeys, ducks, and geese.

prescription

A written order from a health care provider for medicine, therapy, or tests.

prevent

To stop from happening.

protein

A molecule made up of amino acids that the body needs for good health. Proteins are the basis of body structures such as skin and muscle, and substances such as enzymes and antibodies.

proton pump inhibitor

PPI. A drug that reduces the amount of acid made by the stomach. It is used to treat peptic ulcer and gastroesophageal reflux disease.

red blood cell

A cell that carries oxygen to and removes carbon dioxide from all parts of the body.

restless leg syndrome

A disorder of the nervous system that causes unpleasant or painful feelings in the legs, especially when relaxing, which results in uncontrollable urges to move them.

rheumatoid arthritis

An autoimmune disease that causes pain, swelling, and stiffness in the joints. It may cause severe joint damage, loss of function, and disability. The disease may last from months to a lifetime, and symptoms may improve and worsen over time.

risk

The chance or probability that a harmful event will occur. In health, for example, the chance that someone will develop a disease or condition.

spleen

A fist-size organ under your ribs and above your stomach on the left side of your body. Your spleen filters your blood and breaks down old red blood cells, stores blood for emergencies such as trauma, and helps your body fight infection by making white blood cells that destroy bacteria, viruses, and other germs.

supplement

A nutrient that may be added to the diet to increase the intake of that nutrient. Sometimes used to mean dietary supplement.

symptom

A feeling of sickness that an individual can sense, but that cannot be measured by a healthcare professional. Examples include headache, tiredness, stomach ache, depression, and pain.

thyroid

A gland located in the front of the neck, below the larynx (Adam's apple). The thyroid makes hormones that circulate in the bloodstream and affect brain development, metabolism, weight, breathing, heart rate, blood pressure, nervous system functions, body temperature, muscle strength, skin dryness, menstrual cycles, and cholesterol levels.

toxic

Having to do with poison or something harmful to the body. Toxic substances usually cause unwanted health effects.

treat

To care for a patient with a disease by using medicine, surgery, or other approaches.

ulcer

A sore on the skin or in the lining of an organ or other tissue that deepens as cells die. Ulcers form where swelling, redness, pain, and warmth (inflammation) have killed the cells, then bleed and produce pus.

upper limit

UL. The largest daily intake of a nutrient considered safe for most people. Taking more than the UL is not recommended and may be harmful. The UL for each nutrient is set by the Food and Nutrition Board at the National Academies of Sciences, Engineering, and Medicine. For example, the UL for vitamin A is 3,000 micrograms/day. Women who consume more than this amount every day shortly before or during pregnancy have an increased chance (risk) of having a baby with a birth defect. Also called the tolerable upper intake level.

US Department of Agriculture

USDA promotes America's health through food and nutrition, and advances the science of nutrition by monitoring food and nutrient consumption and updating nutrient requirements and food composition data. USDA is responsible for food safety, improving nutrition and health by providing food assistance and nutrition education, expanding markets for agricultural products, managing and protecting US public and private lands, and providing financial programs to improve the economy and quality of rural American life.

vegetarianism

The practice of avoiding all or most animal products for environmental, philosophical, and health reasons. Vegetarians (people who practice vegetarianism) eat a diet based on foods that come from plants and may include some dairy products and eggs. See: vegetarian diet.

vitamin

A nutrient that the body needs in small amounts to function and maintain health. Examples are vitamins A, C, and E.

vitamin C

A nutrient needed by the body to make collagen (a protein found in cartilage, tendons, ligaments, bone, and blood vessels), to absorb iron from food, and for wound healing. It is an antioxidant and protects cells from free radical damage. Vitamin C is found in some foods including citrus fruits, strawberries, peppers, dark green vegetables, tomatoes, and potatoes. Also called ascorbic acid.

zinc

A mineral found in most cells of the body. It helps enzymes work properly, helps maintain a healthy immune system, helps maintain the senses of taste and smell, and is needed for wound healing, making DNA, and normal growth and development during pregnancy, childhood, and adolescence. Zinc is found in some foods, including oysters, red meat, poultry, beans, nuts, certain seafood, whole grains, fortified breakfast cereals, and dairy products.

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